

The Field-Tensor Dictionary: From the Mechanical State of a Tensioned Two-Strand Rope to Electromagnetism

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July 2026. The rigorous assembly of the corpus’s electromagnetic identifications into a single mapping: mechanical state → potentials → field tensor → forces, with every line either Derived or explicitly flagged, and both force-sign rules obtained from mechanics with nothing put in by hand.

1. The mechanical state variables

The two-strand rope under tension T_0 carries one internal gapless degree of freedom: the braid phase Φ , the single Goldstone mode of the internal symmetry (the screw generator; the one-generator theorem, GRV-020). Its derivatives are the physically meaningful densities: the TWIST DENSITY $\eta = \partial\Phi/\partial s$ (structural handedness per unit length) and the ROTATION RATE $\omega = \partial\Phi/\partial t$ (the helix turning like a screw). The tension channel couples to twist through the twist-stretch lock, $k_{tw} = k + \lambda \gamma^2 \tau_0^2$ with $\gamma = 1/\sin^2\theta$ derived from helix geometry (EM-RECON-012). Coarse-graining the strand orientations over many ropes yields the orientation field, carrying a connection in the fiber-bundle sense (GG-001).

2. The dictionary

Electromagnetic object	Mechanical identity	Status	Claims
Charge q	Winding / linking number of the two strands: an integer; handedness is the sign	Derived (quantized)	GG-006, EM-001
Current J	Transported linking: the helix turned like a screw; one turn = one linking unit past each cross-section; $\text{div } J = 0$ is the no-wind-up condition	Derived	EM-008, EM-014
Vector potential A	The coarse-grained orientation connection; gauge freedom = phase-convention arbitrariness (physically: rigid braid-phase relabeling, the GRV-020 symmetry)	Derived (framework)	GG-001, GG-004

Scalar potential φ	The twist-tension channel: stored twist transmitted to tension by the twist-stretch lock	Modeled	EM-RECON-011, -012
$E_i = F_{i0} = -\partial_i \varphi - \partial_t A_i$	Static part: along-rope tension/twist gradient. Induction part: time-varying orientation transport (Faraday, inside $dF = 0$)	Assembled here	EM-RECON-011, EM-003
$B_i = \frac{1}{2} \epsilon_{ijk} F_{jk}$	Curl of the orientation field: the surrounding network's circulating response to the rotation rate ω	Derived / Modeled	EM-009, EM-010, EM-014
Field magnitude	$g = E \sqrt{(\epsilon_0 / \Sigma)}$: strain per field, with ONE constant $\Sigma = T_{on_L}$ (vacuum tension density), bounded by ATLAS/PVLAS	Modeled (calibrated form)	EM-RECON-014, -016
Maxwell equations	$dF = 0$ (Bianchi identity) + $d * F = J$ (Chern-Weil, $d = 3$)	Derived	EM-003
Lorentz force $q(E + v \times B)$	From the coupling $q \cdot v \cdot a$, FORCED by gauge invariance + current conservation; correct coefficient; $\perp$ to v and B ; does no work	Derived	EM-013, EM-010
Magnetic force sign	Gap geometry: same-sense circulations produce opposite swing velocities between wires $\rightarrow$ mesh $\rightarrow$ attract; opposite $\rightarrow$ clash $\rightarrow$ repel	Derived	EM-012
Electrostatic force sign	Constraint-source superposition: like twist fields ADD between knots (repel); opposite CANCEL (attract); $+q_1 q_2 / (4\pi K d)$	Derived (new)	EM-015

3. The electrostatic sign theorem (the new result)

The trap, named first: a quadratic field between sources does NOT automatically repel like charges. With COUPLING sources (an energy term $-\rho\Phi$, the structure of scalar exchange), the minimized interaction is $-q_1q_2/(4\pi Kd)$: like charges attract — which is why scalar-mediated forces attract and why naive arguments get Coulomb backwards. The rope winding is not a coupling source. Winding number is TOPOLOGY: a boundary condition on the braid-phase configuration space, with nothing to couple and nothing to switch off. For such CONSTRAINT sources the energy is purely $(K/2)|\nabla\Phi|^2$ on the constrained minimizer, and superposition gives $U = +q_1q_2/(4\pi Kd)$: like windings repel, opposite attract, with the $1/d$ potential and $1/d^2$ force of Coulomb's law. The laboratory precedent is exact: superfluid vortices are constraint-sourced quadratic fields, and like-sign vortices repel while vortex–antivortex pairs attract. Electron and positron are opposite handedness (GG-006): opposite winding, hence attraction. The benchmark verifies both the analytic dichotomy (the two source types provably carry opposite signs, so the trap is permanently fenced) and the mechanism on a lattice of actual winding defects: the like pair's energy falls with separation (repulsion), the opposite pair's rises (attraction). In plain terms: like-handed twist fields add in the region between the knots, storing more energy when they are close, so they are pushed apart; opposite-handed fields cancel between the knots, so they are pulled together. With the magnetic half (the gap-geometry rule of EM-012), the sign ledger of electromagnetism is now closed entirely by geometry.

4. The status ledger: why Modeled and not Derived

The dictionary is assembled with its remaining debts explicit. (i) Σ 's absolute value is an input: measured and bounded by experiment (ATLAS; PVLAS), identified in form as T_{on_L} , not derived from within. (ii) The dynamical drive's inertial term (EM-010) is assumed, not derived. (iii) The scalar potential's channel identification (EM-RECON-011) is Modeled. (iv) The identification's UNIQUENESS — that no other assignment of mechanical variables reproduces the same phenomenology — is not established. On this assembly the corpus's central identification claim, GG-005 ('F IS the electromagnetic field; topology IS charge'), is upgraded from Conjecture to Modeled: supported by structure, quantization, forces with both signs, calibration, and dynamics; withheld from Derived by exactly the four debts above.

5. What would strengthen each line

Deriving Σ from the mesh scale and tension (blocked at present by the same irreducibility results that block a and $\hbar$); deriving EM-010's inertial term from strand dynamics; promoting the φ -channel identification by an independent prediction in the static sector; and a uniqueness argument — or a counterexample assignment, which would be equally valuable and would go into the registry the same way everything else has.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **EM-015 [Derived]:** ELECTROSTATIC SIGN THEOREM [benchmark: benchmarks/em/electrostatic_sign_derivation.py]
- **EM-015 [Derived]:** ELECTROSTATIC SIGN THEOREM [benchmark: benchmarks/em/electrostatic_sign_derivation.py]
- **EM-016 [Modeled]:** THE FIELD-TENSOR DICTIONARY [benchmark: benchmarks/em/electrostatic_sign_derivation.py]
- **EM-016 [Modeled]:** THE FIELD-TENSOR DICTIONARY [benchmark: benchmarks/em/electrostatic_sign_derivation.py]